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Solution

Quiz 2

Q1: Consider a scenario where you are connected with any residential network as an end system with LAN. Residential router has an access link rate of 2.54 Mbps. RTT from residential router to Server is 2 secs. Now you are generating a HTTP request from browser to server with average object size of 350 Kbits. Number of average requests sent from your browser is 7 requests/sec. Find average access delay on this network. **(5 Marks)**

Q2: Suppose there is an institutional network connected to the Internet having average object size of 850,000 bits and that the average request rate from the institution's browsers to the origin servers is 12 requests per second. Also suppose that the amount of time it takes from when the router on the Internet side of the access link forwards an HTTP request until it receives the response is three seconds on average.
Model the total average response time as the sum of the average access delay (that is, the delay from Internet router to institution router) and the average Internet delay.

[1] Find the total average response time. **(3 Marks)**
[2] Now suppose a cache is installed in the institutional LAN. Suppose the miss rate is 0.4. Find the total response time. **(2 Marks)**

*Access link rate 15 Mbps
*LAN rate 100 Mbps

Q1 $R/C = 2540000$ $L = 350000$ req/s $= 7$ RTT $= 2s$

① Trans time $= L/R = \frac{350000}{2540000} = 0.138\ s$

② max. serving ability $= 1/0.138 = 7.26$ req/s

③ utilization: $7/7.26 = (96.5\%)$

④ avg access delay $= \frac{1}{\text{max serv. ability} - \text{req}} = \frac{1}{0.26} = 3.84$ sec

Total response time $= 3.84 + 2 = 5.84$.

Q2 ① $\frac{850000}{15 \times 10^6} = 0.056667$

$12 \times 0.056667 = 0.680004$

$\frac{0.05667}{1 - 0.680004} = 0.177$

$+3 = 3.177$ sec

② Cache

miss $= 0.4 \times 3.1777 = 1.2708$

hit $= \frac{850000}{100 \times 10^6} = 0.0085 \rightarrow 12 \times 0.0085 = 0.102$

$\frac{0.0085}{1 - 0.102} = 0.0094$

total $= 1.2708 + \underbrace{(0.6)(0.0094)}_{\text{hit}} \rightarrow$ total $= 1.2714$.